Plant and Animal Tissues

K	N	L
I know what are Tissues	I want to learn about ciliated epidermis

2. Keywords

1. Merestamatic Tissues
2. Epidermis
3. adipose tissues
4. parenchyma
5. tracheid
6. arcolar tissue
7. ciliated apethelium
8. red blood capsules
9. collenchyma
10. sclerenchyma
11. Columner epithelium
12. cuboidal epithelium.
13. squamous epithelium
14. nervous tissues
15. pauession canals.

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Assignment - 1

Short answer questions.
Define the term given below.

i) Tissue - A group of cells, which are similar in structure and perform a specific function, form a tissue.

ii) Organ - A specialized structure made up of different types of tissues that work together to perform a specific function.

iii) Organ System - A group of individual organs that work together as a team to perform a function.

II Define the following:

1) Tissue - A group of cells similar in structure that performs a specific function.

2) organ - Diffrent tissues working together to perform a specific function

3) organ System - A group of organs working togeth to perform a specific function

IV Answer the following questions.

1. Distinguish between the following pair on the basis of the words indicated in the Brackets.

a) Simple perment tissue and complex permenent tissue (cellular organisation)

Simple permenent ~~tissue~~	Complex permenent tissue
* Made up of only one type of cell.	* Made up of more than one type of cell.
* All cells are similar in struct	* Cells are diffrent in structure,
* Cell perform same basics. Ex - Parenchyma, collenchyma.	* Diffrent cells work together to perform common function Ex - xylem, phloem

Assignment -2

1) why are meresmetic cells found mainly at the tip of the roots and shoots

Ans Merestamatic tissues are diving tissues which help in the growth of the plant.

2) Why can plant's regrow new plant beigns but animals cant do so in the same way.

Ans Plants can regrow themselves because they consist of merestematic tissues which are also called dividing tissues. These tissues are not present in animals so they do not regrow.

Q. A plants phloem tissues was damaged due to a disease. what effect would this have on the plant and why?

Ans If the phloem tissue is changed, damaged, food prepared by the leaves through photosynthesis cannot be transportal to the roots, stems and other parts of the plant. As a result, all parts except the leaves would be deprived of food and would slowly weaken and die. Eventually the whole plant could die as food supply is completly cut off.

Q3 A dry desert plant has very thick and rigid stems that do not bend even in strong winds. Which type of permenent tissue is most likely responsible for this and why?

Ans Sclerenchyma tissue is most likely responsible for this sclerenchyma cells have very thick cell wall that make them extremely rigid and strong. These dead cell provides maximum mechanical support and rigidity to the plant body, which helps the desert plant with stand strong winds without bending or breaking.

Assignment - 3

Xylem cells are dead at maturity which phloem cells are alive. A student says dead cells cannot perform any functions So xylem is important than phloem Do you agree justify your answer.

Ans No, I do not agree. Both xylem and phloem are essential for a plant's survival, and the dead state of xylem cells is actually a functunal advantage.

Function of Dead

In xylem cells like tracheids and vessels lose their protoplasm at maturity to become hollow tribes.

Structural support

The thick lignified walls of dead xylem cells provide the chemical strenght neccesary to keep the plant upright

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without xylem, the plant would dehydrate and collaspe and the plant would starve.

4. Give four differenciate between bone and cartilage.

Bone	Cartilage
• Hard and Rigid	• Soft and elastic
• Non-flexible and porous.	flexible and non-porous
• Contains osteocytes	contains chondrocytes
• Blood vessels present.	Blood vessels present.

5 Give reason for the following:

1) The trachea is covered by cartilage rings and not tough bone.

Ans Cartilagious rings are present in the trachea is trachea to prevent its collaspe when air is not passing through it. These rings are made of flexible connective tissue that support the trachea while still allowing it to move and bend.

2. A bone is a hard tissue.

Ans Bone is primarly composed of a hard matrix that includes cotecytes, which are mature bone cells embedded within it. The matrix is rich in minerals, particularly calcium and phosphurus salts, which contribute to the hardness and strenght of the bone.

3. A tissue sample was observed under a microscope. the cells were tightly packed with no intercellular spaces and formed a continous layer. Identify the tissue where in the body is it most likely found? what is its function?

Ans This is epthical tissue. It is most likely found lining the skin digestive tract, respiratory tract respiratory track and other body surfaces. It prevents entry of germs, dust and harmful substances into the body. It also helps in absortion and secretion.

4) Mention the specific functions of each of the following tissues.

Ans i) Areolar tissue - It binds our skin to the underlying tissues.

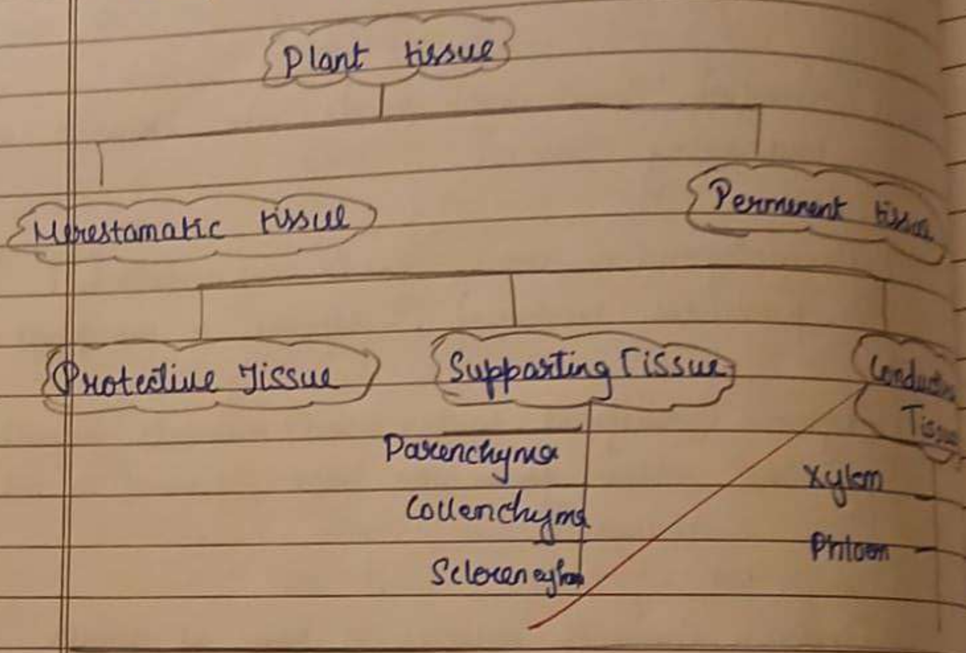
ii) Parenchyma - Stores food materials and provides temprary support to the plant.

iii) Xylem - Transport water and minerals absorbed by the roots from the soil upward to the leaves.

iv Tendon - Connects muscles to bone at joints.

v) Adipose tissue - Store fat in the body - It acts as an insulation and keeps the body warm. It also protects the internal organs by acting as a cusion around them.

Concept Map


Plant tissue
Merestamatic tissue
Permenent tissue
Protective Tissue
Supporting Tissue
Conducting Tissue
Parenchyma
Collenchyma
Sclerenchyma
Xylem
Phloem

Animal Tissue

Epthical Tissue

Squamous epithilium

Cuboidal epithilium

Columnar epithilium

Ciliated epithilium

Connective

fibrous

connective

Bone

Cartila

fluid - blood
lymph

Muscular

nervous

neurons

Areolar
Adipose
tendons
ligament

Neat work

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Snap test

1 Arrange the terms in each group in the correct order. Write them in a logical sequence beginning with the term that that is underlined.

1) organism , cell , organ , Tissue , Organ system

2) Plant growth, Merestamatic cells , Formation of permanant tissue , Division of cells.

3) water reaches leaves , Water movement in stem Root absorption , Transpiration

4) Distribution to other parts , Food manufactured in leaves Storage in roots / leaves, Transport through phloem.

5) Age estimation of plants , formation of a new new xylem layer, Activity of cambium Development of annular ring.

Ans 1) cell tissue, organs, organ system, organism

2) merestamatic tissue, Division of cells,
formation of peramenent tissue, plant growhth.

3) food manufactured in leaves,
distribution to other parts,
fruits / roots, Transport through phloem

4) Age estimation of plant,
Formation of new xylem layer.

Extra lines

1) what is haversian canal ?

Ans The heaversian canal is the central canal in compact bone that contains blood vessels and nerves.

2) What is neurotransmitter ?

Ans A neurotransmitter is a checimal messenger that carries impulsers from one nerve cell to another or farm a nerve cell to a muscle or glad.